Members

Column	Type	Key	Nullable
member_id	INT	PK	no
first_name	VARCHAR(100)		no
last_name	VARCHAR(100)		no
email	VARCHAR(255)	UNIQUE	no
zip	CHAR(5)		no
joined	DATE		no
status	ENUM(‘active’, ‘suspended’)		no

Staff

staff_id	INT	PK	no
first_name	VARCHAR(100)		no
last_name	VARCHAR(100)		no
email	VARCHAR(255)	UNIQUE	no

Categories

category_id	INT	PK	no
name	VARCHAR(100)		no
daily_late_fee	DECIMAL(8,2)		no
max_loan_days	INT		no
requires_cert	BOOLEAN		no

Tools

tool_id	INT	PK	no
name	VARCHAR(150)		no
brand	VARCHAR(100)		no
category_id	INT	FK -> categories(category_id)	no

purchased	DATE		no
price	DECIMAL(10,2)		no
condition	ENUM('GOOD', 'FAIR', 'NEEDS_REPAIR', 'RETIRED')		no

Certifications

member_id	INT	PK, FK->members(member_id)	
category_id	INT	PK, FK->categories(category_id)	
certified_on	DATE		no

Loans

loan_id	INT	PK	no
tool_id	INT	FK->tools(tool_id)	no
member_id	INT	FK->members(member_id)	no
staff_id	INT	FK->staff(staff_id)	no
out_date	DATE		no
due_date	DATE		no
back_date	DATE		no
fee_charged	DECIMAL(8,2)		yes

Q1: Relation: loans(loan_id, member_id, staff_id, out_date, due_date, back_date, fee_charged)
Functional dependencies:

- Loan_id → tool_id, member_id, staff_id, out_date, due_date, back_date, fee_charged

Candidate key: (loan_id)
Why is it in 3NF? There is only one non-trivial functional dependency, which has the candidate key loan_id on its left. This means the non-prime attributes depend directly on the primary key. There are no partial or transitive dependencies.

Q2: Relation: certifications(member_id, category_id, certified_on)
Functional dependencies:

- Member_id, category_id → certified_on

Candidate key: (member_id, category_id)
Why is it in 3NF? There is also one non-trivial dependency that has the composite candidate key on its left. There are no partial or transitive dependencies.

Q3: The old loan log has repetitions of member details. Member_name is in every loan row, which makes a dependency of member_email to member name. Since email was not a superkey, the loan log contains a transitive dependency, which violates 3NF properties. New loans have member_id as a FK, and member name is displayed once in members. This makes updating Carlos's name happen in only one row. Loans can see the corrected name through a FK join.